

Split-Attention Synchronised Agents: A Self-Contained Calculus of a Bounded Character That Divides Its Attention Across Concurrent Scenes and Synchronises With Others, with an Implementation Blueprint

Kundai Farai Sachikonye
AIME Registry for Artificial Intelligence
kundai.sachikonye@bitspark.com

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Abstract

A useful synthetic character is not confined to one exchange: it persists across sessions, is engaged from several places at once, and must remain recognisably itself while it divides a finite capacity among concurrent demands and, where its ends overlap with others', coordinates with them. We give a self-contained mathematical account of such a *split-attention synchronised agent* and derive, as theorems from stated axioms, both its behaviour and the exact commitments an implementation must honour. We model an agent as a finite weighted graph — its *self-graph* — carrying a conserved invariant we compute from its own internal boundary content, a purpose that is the attractor of a strongly convex drive, and a bounded attention budget. A *scene* is an outward context in which the agent may act; the agent faces many scenes at once and must apportion attention among them. From five structural axioms on the agent (finiteness, presence, costly individuation, a positive floor on the cost of any act, and phase exclusion), plus one clearly-marked environmental assumption (diminishing returns of attention within a scene), we prove: **(T1)** the agent carries a conserved, non-local, strictly positive identity invariant, fixed under every relabelling of its parts, so a character is the same character across scenes and across sessions; **(T2)** in a diminishing-returns environment, optimal division of attention over concurrent scenes is a

water-filling rule with a single shadow price, so a busy agent is *present but preoccupied* in every scene at once — a guarantee we flag as conditional on the environment, not on the agent; **(T3)** construction and commitment cannot occupy one instant, so an agent that ever reflects is unresponsive on a positive fraction of instants in each scene, a second and independent cause of split appearance; **(T4)** every committed act deposits an unrepealable increment in a monotone count, so an agent never re-occupies a past state, sessions accumulate rather than reset, and an authored copy is always a distinct individual; **(T5)** recalling is searching — no reply is fetched from store, each is freshly walked over the agent's own graph — so a character stays in character without being reminded to; **(T6)** agents with wholly disjoint self-graphs synchronise on a shared purpose defined in a common outcome space, at a coordination cost that vanishes exactly when their outward phases lock — and we supply the mean-field coupling dynamics that *drive* dispersed phases to lock above a computable threshold, so lock is achieved, not assumed — and a crowd of them attains the shared purpose more sharply than any one; **(T7)** the society has its own purpose — the attractor of its own strongly convex drive — that maps to the shared purpose in outcome space, so synchronisation among split-attention agents is itself a scene-allocation problem one level up, and a society divides its collective attention by the same water-filling law as a single agent be-

cause the same purpose-as-attractor invariant propagates across the quotient from agents to society. We then convert the theorems into an *implementation blueprint*: four invariants a run-time must preserve — a computable conserved invariant, a never-resetting committed count, search-not-fetch world access, and exclusive construction/commitment phases — each stated as a checkable predicate with the theorem that certifies it, and we separate the guarantees that hold unconditionally from the agent’s structure from the one (optimal attention division) that is conditional on the environmental assumption. We give the constructions, the water-filling and synchronisation algorithms, and a numerical study on explicitly built agents. Throughout, the interpretive readings of the formal objects — “identity,” “attention,” “distraction,” “coordination” — are marked as orientation and confined to remarks; no theorem depends on them.

Keywords: bounded agent; conserved invariant; attention allocation; water-filling; phase exclusion; monotone history; recognition as search; synchronisation; outcome-space coordination; implementation invariant; non-player character.

1 Introduction

1.1 The object

Consider a synthetic character that is not summoned for a single question and dismissed. It runs. It is addressed from more than one place at a time — one context asks it to do one thing while another asks it for something else — and it is met again tomorrow, expecting to find the same character it left. Such an agent must do three things at once that a one-shot responder never has to. It must *remain itself* across all the contexts it is engaged from and across the gap between sessions. It must *divide* a capacity it cannot enlarge among the several contexts pulling on it, so that it is fully committed to none and wholly absent from none. And where its own ends happen to line up with those of other such characters, it must *coordinate* with them without first being given a shared mind.

This paper asks what those three requirements *are*, formally, and what a system that meets them must do. We want definitions sharp

enough to compute with, general enough to bind any bounded agent, and self-contained: the reader should need nothing but this paper and standard mathematics. And because the intended use of the account is to build the object it describes, we want each behavioural result paired with the precise commitment a running system must honour to exhibit it.

1.2 The three requirements, and how the paper meets them

Remaining itself. We model the agent as a finite weighted graph whose vertices are the distinctions it draws within itself and whose edge weights are the costs of holding them apart, and we extract a quantity — the minimum internal boundary content — that is conserved under every relabelling of the agent’s parts. This quantity is positive, is spread across the agent rather than located in any one part, and cannot be resolved to a point from outside. It is the standing structure from which the agent’s changing conduct derives (Section 4).

Dividing attention. We give the agent a finite attention budget and a family of concurrent *scenes*, each offering a purpose gain per unit of attention spent in it. We prove that the value-maximising division of attention is a water-filling rule governed by a single shadow price, and that this makes the agent present-but-preoccupied in every scene simultaneously, with a readable dial for how reachable it is (Section 5). A second, independent cause of divided appearance is that forming new distinctions and committing to an act cannot share an instant (Section 6).

Coordinating. We show a society of such agents is again a graph, so the identity apparatus applies one level up; that agents with wholly disjoint internals can nonetheless attain a shared purpose defined in a common space of outcomes; that the cost of coordination vanishes exactly when their outward phases lock; and that a crowd sharpens a shared purpose (Section 8). Synchronising a society turns out to be a scene allocation problem one level up, closing the construction (Section 9).

1.3 Blueprint, not metaphor

Section 10 is the paper’s purpose made operational. We isolate four *implementation invariants* — a computable conserved invariant, a never-resetting committed count, search-not-fetch world access, and exclusive construction/commitment phases — and for each we state the predicate a runtime must satisfy and name the theorem that makes the predicate the right one. A system honouring the four invariants exhibits the behaviour derived here; a system violating any one degrades into a locally-plausible responder that resets. The invariants are design commitments, honourable by construction, not empirical hypotheses about any existing system.

1.4 What we do and do not claim

We do not claim that human or animal identity, attention, or coordination is a finite graph or a convex program. We claim that *any* bounded agent satisfying the five structural axioms of Section 2 has the identity, history, phase, and coordination structure we derive, and — where its environment additionally offers diminishing returns (the sixth, environmental axiom) — the water-filling attention division; and that this structure suffices to found a usable notion of a persistent, attention-dividing, coordinating synthetic character. The readings of the formal objects as “identity,” “attention,” “distraction,” “coordination,” and “character” are offered as orientation and confined to remarks; the theorems are graph-theoretic, order-theoretic, and convex-analytic, and are used only as such. We prove a conditional and then note that, for synthetic agents, its antecedent is something a builder may simply arrange to hold.

1.5 Organisation

Section 2 fixes the agent and states the axioms. Section 3 proves the floor. Section 4 derives the conserved identity invariant (T1). Section 5 proves the water-filling attention division (T2). Section 6 proves phase exclusion (T3). Section 7 proves the monotone, irreversible history and the distinctness of copies (T4) and that recalling is searching (T5). Section 8 places agents in society and proves disjoint coordina-

tion, phase-locked zero-cost synchronisation — together with the mean-field coupling dynamics that drive phases to lock above a threshold — and crowd sharpening (T6). Section 9 gives the society its own purpose-as-attractor and proves society-level attention division as genuine invariant propagation (T7). Section 10 states the four implementation invariants and the algorithms, separating unconditional from environment-conditional guarantees. Section 11 reports a numerical study with exact, cross-method, Kuramoto-integrated, and negative-control checks. Section 12 discusses scope.

2 The agent, the scene, and the axioms

We build the agent from the inside as a graph of internal distinctions, give it a disposition space and a drive whose bottom is its purpose, and place it before many scenes at once.

2.1 The self-graph

Definition 2.1 (Self-graph). The *self-graph* of an agent A is a finite weighted graph $\Gamma = (V, E, w)$ where V is a finite set of *parts* (the internal distinctions the agent draws), $E \subseteq \binom{V}{2}$ is a set of *separations* (pairs of parts held apart), and $w : E \rightarrow (0, \infty)$ assigns to each separation the *cost* of holding it. The graph is simple and, by Axiom 2.7, connected.

Definition 2.2 (Cut and boundary). For disjoint $S, T \subseteq V$, the *cut* between them is $\partial(S, T) = \{uv \in E : u \in S, v \in T\}$ with weight $w(\partial(S, T)) = \sum_{e \in \partial(S, T)} w(e)$. For $\emptyset \neq U \subsetneq V$ the *boundary* of U is $\partial(U) = \partial(U, V \setminus U)$ with *boundary cost* $b(U) = w(\partial(U))$.

2.2 Dispositions, drive, purpose

Definition 2.3 (Disposition space; drive; purpose). The *disposition space* $\mathcal{B}$ of an agent is a compact convex subset of $\mathbb{R}^n$; a point $x \in \mathcal{B}$ records the agent’s current tendencies to act. A *drive potential* is a continuously differentiable $\Phi : \mathcal{B} \rightarrow \mathbb{R}$ that is λ -strongly convex for some $\lambda > 0$:

$$\Phi(y) \geq \Phi(x) + \nabla \Phi(x)^\top (y - x) + \frac{\lambda}{2} \|y - x\|^2, \quad \forall x, y \in \mathcal{B}.$$

The *purpose* is the unique minimiser $x^* = \arg \min_{x \in \mathcal{B}} \Phi(x)$; the *residual to purpose* at x is $D(x) = \Phi(x) - \Phi(x^*) \geq 0$.

2.3 Scenes

The novelty of the present setting is that the agent faces *several* outward contexts at once. We call each a scene.

Definition 2.4 (Scene; scene field). A *scene* S presented to the agent is an outward context in which the agent may spend attention to advance its purpose: formally a pair $S = (A_S, \gamma_S)$ where $A_S : \mathcal{B} \rightarrow \mathcal{O}$ maps the agent’s disposition to an outcome in a common *outcome space* $\mathcal{O}$, and $\gamma_S : [0, \infty) \rightarrow [0, \infty)$ is the scene’s *gain profile*: $\gamma_S(a)$ is the purpose residual the agent can remove by committing attention a in scene S . Each γ_S is nondecreasing and continuously differentiable with $\gamma_S(0) = 0$; its concavity is *not* part of this definition but is imposed separately, where needed, by Axiom 2.11. The agent’s *scene set* at an instant is a finite family $\mathcal{S} = \{S_1, \dots, S_k\}$.

Remark 2.5 (Concavity is a property of the environment, not the agent). Concavity of γ_S (Axiom 2.11) says the first unit of attention in a scene removes at least as much residual as the next: attention has diminishing returns within a scene. It is what makes the attention-division problem convex and gives the water-filling law of Section 5. But it is crucial to be clear about its status: it is a fact about the *world*, not a theorem about the agent. Nothing in the agent’s structure — its self-graph, its drive, its budget — forces the scenes it faces to reward attention with diminishing returns. An environment can perfectly well present a scene of *increasing* returns over some range: a task where the first sliver of attention accomplishes nothing and only sustained concentration pays off (a lock that opens only when fully worked, a proof that yields only when pursued past a threshold). For such non-concave scenes the division problem is not convex and water-filling is *not* optimal: the value-maximising allocation may concentrate the whole budget on one scene rather than spread it by a single price. We therefore state concavity as the separate environmental assumption it is, invoke it only in Theorems 5.2 and 9.5, and, in the blueprint (Section 10), keep the guarantees that depend

on it distinct from those that do not. A linear profile is the boundary case and is admitted.

2.4 The axioms

We state six axioms. The first five are meant to be nearly undeniable for anything we would call a bounded agent, and — the point of Section 10 — to be arrangeable by construction in a synthetic one. The sixth is of a different character: it is an assumption about the agent’s *environment*, not about the agent, and we mark it as such where it is used.

Axiom 2.6 (Finiteness). The self-graph is finite: $|V| < \infty$, and the scene set is finite: $|\mathcal{S}| < \infty$.

Axiom 2.7 (Presence). At every instant the agent occupies exactly one part; the self-graph is connected and nonempty. The agent is always somewhere in itself.

Axiom 2.8 (Costly individuation). No internal distinction is free: there is $\beta > 0$ with $w(e) \geq \beta$ for all $e \in E$.

Axiom 2.9 (Act floor and bounded attention). No act is free: every committed act costs at least $\beta_a > 0$, and the agent has a finite *attention budget* $\alpha > 0$ per unit time — the total attention it may commit across all scenes in a unit interval satisfies $\sum_{S \in \mathcal{S}} a_S \leq \alpha$.

Axiom 2.10 (Phase exclusion). Forming a new internal distinction and committing to an act draw on the same instantaneous capacity and cannot both occur at one instant: at each instant the agent is in exactly one of a *construction phase* (it reshapes its distinctions, commits no act) or a *commitment phase* (it commits an act, forms no new distinction).

Axiom 2.11 (Diminishing returns of attention — an environmental assumption). Each scene’s gain profile γ_S (Definition 2.4) is nondecreasing and concave: for a scene S and $0 \leq a_1 \leq a_2$, $\gamma_S(a_2) - \gamma_S(a_1)$ is nonincreasing in the base point, so the first unit of attention removes at least as much purpose residual as any later unit. *This is an assumption about the world the agent acts in, not a consequence of the agent’s structure.* It says the environment offers diminishing, never increasing, returns to attention within any one scene. Unlike Axioms 2.6

to 2.10, which constrain the agent and can be honoured by construction in a synthetic one, Axiom 2.11 constrains the scenes the agent is placed before and can fail in an environment that rewards concentrated attention. We invoke it only where named, and we track in Section 10 exactly which guarantees depend on it.

Remark 2.12 (Costly individuation, motivated). Axiom 2.8 may be motivated rather than merely posited. To individuate a part U is to tell it apart from the rest $V \setminus U$. If distinguishing a part from the rest could be *finished*, the rest would be an exhausted catalogue and the agent would be nothing held in reserve. An agent always in some state (Axiom 2.7) that never reduces to a closed catalogue leaves on each part an irreducible remainder of the comparison against the rest; that remainder is the positive floor. We take Axiom 2.8 as an axiom and prove from it, but the floor is the natural, not an arbitrary, hypothesis.

Remark 2.13 (The act floor is earned, not stipulated). The positive floor β_a on the cost of an act (Axiom 2.9) is likewise not an arbitrary constant but a consequence of what it is to *act*. An act is a commitment: it deposits an increment in the agent’s history (Definition 7.1) that distinguishes the post-act state from the pre-act one. A putative act of zero cost deposits nothing — it leaves the committed count, the disposition, and every separation exactly as they were — and so effects no distinction between before and after. But an operation that changes nothing is not a commitment; it is a no-op that did not occur. Hence every genuine act deposits a positive increment, and the least such increment is a floor $\beta_a > 0$: a zero-cost act is a contradiction in terms, not a cheap act. This is the same argument, one level up, as Remark 2.12: just as no distinction is free because a free distinction distinguishes nothing, no act is free because a free act commits nothing. We take Axiom 2.9 as an axiom and prove from it, but the act floor, like the individuation floor, is forced by the very notion it bounds.

Example 2.14 (A three-part agent facing two scenes). Let $V = \{a, b, c\}$ with separations ab, bc, ca of costs 2, 3, 2, so $\beta = 2$. Let $\mathcal{B} = [0, 1]^3$, $\Phi(x) = \frac{1}{2}\|x - x^*\|^2$ with $x^* = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ ($\lambda = 1$). Let two scenes have gain profiles

$\gamma_1(a) = \ln(1 + 2a)$, $\gamma_2(a) = \ln(1 + a)$, and $\alpha = 1$. We return to this agent throughout.

3 The floor

Theorem 3.1 (Floor). *Under Axioms 2.6 to 2.8, every nonempty proper part $U \subsetneq V$ has boundary cost $b(U) = w(\partial(U)) \geq \beta > 0$. No part is separated from the rest at zero cost.*

Proof. By Axiom 2.7 the graph is connected, so a nonempty proper U has at least one edge to $V \setminus U$; else U and its complement are disconnected. Hence $\partial(U) \neq \emptyset$ and $b(U) = \sum_{e \in \partial(U)} w(e) \geq \beta$ by Axiom 2.8. $\square$

Corollary 3.2 (Realised floor). $\beta_\Gamma := \min_{\emptyset \neq U \subsetneq V} b(U)$ is strictly positive and attained, by finiteness of V .

4 Identity: a conserved invariant across scenes and sessions

We isolate what stays fixed in the agent while its surface conduct and its outward scenes change. The surface is the labelling of parts; a re-description that preserves which parts are separated, and at what cost, changes nothing real. We find the quantity surviving all such relabelling.

Definition 4.1 (Isomorphism; invariant). A *weighted isomorphism* $\varphi : \Gamma \rightarrow \Gamma'$ is a bijection $\varphi : V \rightarrow V'$ with $uv \in E \iff \varphi(u)\varphi(v) \in E'$ and $w'(\varphi(u)\varphi(v)) = w(uv)$. A quantity ι is an *invariant* if $\iota(\Gamma) = \iota(\Gamma')$ whenever $\Gamma \cong \Gamma'$.

Definition 4.2 (Internal residual; character invariant). For a partition $\mathcal{Q} = \{U_1, \dots, U_r\}$ of V into $r \geq 2$ nonempty blocks, the *internal residual* is $\rho(\mathcal{Q}) = \sum_{i < j} w(\partial(U_i, U_j))$. The *character invariant* of $\mathbf{A}$ is

$$\chi(\mathbf{A}) = \min_{\mathcal{Q}: r \geq 2} \rho(\mathcal{Q}),$$

attained by finiteness of V : the least total cost of splitting the agent into pieces at all, in the currency of its own separations.

Theorem 4.3 (Identity is a conserved, positive, non-local region). *Under Axioms 2.6 to 2.8:*

(i) **Positive:** $\chi(A) \geq \beta > 0$.

(ii) **Conserved:** χ is a weighted-graph invariant; in particular it is fixed by every relabelling of the agent's parts, and does not depend on which scene the agent is engaged from.

(iii) **Non-local:** χ is in general realised by a partition into multi-part blocks, never by isolating a single part; it is a region, not a point.

Proof. (i) A partition with $r \geq 2$ has, by connectedness, a crossing edge of weight $\geq \beta$, so $\rho(Q) \geq \beta$; the minimum inherits the bound. (ii) For a partition Q of V , its image $\varphi(Q)$ is a partition of V' , and since φ preserves edges and weights, $\rho(\varphi(Q)) = \rho(Q)$; as $Q \mapsto \varphi(Q)$ bijects nontrivial partitions (inverse φ^{-1}), the minima agree. The definition of χ refers only to Γ , not to any scene A_S , so scene engagement leaves it fixed. (iii) Two triangles of edge cost 2 joined by one edge of cost $\beta = 2$: cutting the joining edge separates them at cost 2; isolating any vertex costs $\geq 2 + 2 = 4$. The minimum is the two-block partition, so the invariant is non-local. $\square$

Remark 4.4 (Interpretation: the same character everywhere, on which nothing depends). Theorem 4.3 licenses reading $\chi(A)$ as the agent's standing *identity*: conserved under all re-description, positive, spread across the agent, and — crucially for the present setting — *independent of the scene*. A character engaged from three contexts at once presents the same conserved interior in all three, because the invariant is a property of Γ and the scenes are maps out of $\mathcal{B}$, not alterations of Γ . We use only the graph statement.

Corollary 4.5 (Character is not a state). *No single part carries the character: on the two-triangle agent no singleton $\{v\}$ has $b(\{v\}) = \chi(A)$. Identity cannot be read off any momentary state; it is the standing relation among all of them.*

5 Attention division: water-filling over concurrent scenes

The agent faces $\mathcal{S} = \{S_1, \dots, S_k\}$ at once and cannot attend to all fully. We prove the value-maximising division of a finite budget is a water-filling rule with a single price, and read off the present-but-preoccupied phenomenology.

Definition 5.1 (Attention allocation; value). An *attention allocation* is a vector $a = (a_1, \dots, a_k) \in [0, \infty)^k$ with $\sum_i a_i \leq \alpha$. Its *value* is the total purpose residual removed, $G(a) = \sum_{i=1}^k \gamma_i(a_i)$.

Theorem 5.2 (Optimal attention division is water-filling). *Under Axiom 2.9 and the environmental assumption Axiom 2.11 (concave gain profiles), the value-maximising allocation solves $\max_{a \geq 0, \sum_i a_i \leq \alpha} \sum_i \gamma_i(a_i)$, and there is a single attention price $p^* \geq 0$ (the Lagrange multiplier of the budget) such that the optimal a_i^* satisfies*

$$\gamma'_i(a_i^*) = p^* \text{ if } a_i^* > 0, \quad \gamma'_i(0) \leq p^* \text{ if } a_i^* = 0,$$

i.e. the agent equalises marginal purpose gain across all attended scenes at the price p^ , and drops any scene whose first unit of attention returns less than p^* . The price p^* is nonincreasing in the budget α and nondecreasing in the number and richness of competing scenes. If $\sum_i a_i^* < \alpha$ then $p^* = 0$.*

Proof. The objective is concave (sum of concave γ_i) and the feasible set $\{a \geq 0 : \sum_i a_i \leq \alpha\}$ is compact convex, so a maximiser exists; strict concavity where $\gamma''_i < 0$ makes each active a_i^* unique. Form the Lagrangian $L = \sum_i \gamma_i(a_i) - p^*(\sum_i a_i - \alpha) + \sum_i \mu_i a_i$ with $p^* \geq 0$, $\mu_i \geq 0$. The KKT stationarity condition is $\gamma'_i(a_i) - p^* + \mu_i = 0$ with complementary slackness $\mu_i a_i = 0$ and $p^*(\sum_i a_i - \alpha) = 0$. For $a_i^* > 0$, $\mu_i = 0$ gives $\gamma'_i(a_i^*) = p^*$; for $a_i^* = 0$, $\mu_i \geq 0$ gives $\gamma'_i(0) = p^* - \mu_i \leq p^*$. Sufficiency of KKT for the global optimum follows from concavity. Budget monotonicity: relaxing α enlarges the feasible set, so the optimal value is nondecreasing and its marginal $p^* = \partial \text{val} / \partial \alpha$ is nonincreasing by concavity of the value in α . Adding a scene or raising a γ_i can only tighten the binding budget, raising the level at which margins equalise. If the budget is not exhausted at the optimum, its multiplier is zero. $\square$

Corollary 5.3 (Present but preoccupied in every scene at once). *At the optimum the agent spends positive attention in exactly the scenes whose marginal gain at zero exceeds p^* , and none in the rest; within attended scenes it holds every marginal gain equal to the common price p^* . Thus a busy agent (high p^*) is thinly but nonzero-present in its high-value scenes and absent from low-value ones, in all scenes simultaneously, and the scalar p^* is a single readable dial for how reachable the character is at that moment. Lowering p^* (an idle agent, ample budget) makes it forthcoming everywhere at once.*

Proof. Immediate from the KKT characterisation of Theorem 5.2: the support of a^* is $\{i : \gamma'_i(0) > p^*\}$, margins on the support equal p^* , and p^* falls as α rises. $\square$

Corollary 5.4 (A token presence is the boundary case). *A scene with $\gamma'_i(0)$ only marginally above p^* receives a floor-sized allocation $a_i^* \geq 0$ close to zero — the formal minimal acknowledgement: the agent spends a sliver of attention to remain nonzero-present in a scene it cannot afford to engage fully.*

Proof. As $\gamma'_i(0) \downarrow p^*$, the active-scene condition $\gamma'_i(a_i^*) = p^*$ forces $a_i^* \downarrow 0$ by continuity of γ'_i ; the allocation is positive but arbitrarily small. $\square$

Remark 5.5 (Interpretation: split attention, derived not stipulated). Corollary 5.3 is the paper’s title phenomenon obtained as a theorem. An agent does not need a “distraction module” to appear divided across the several contexts engaging it: dividing a finite budget optimally over concurrent concave demands *is* water-filling, and water-filling is exactly presence-everywhere-preoccupied-everywhere with a single price. The appearance of split attention is what optimal attention looks like from any one scene.

Remark 5.6 (Where the water-filling law can fail). Theorem 5.2 is exactly as strong as Axiom 2.11, and no stronger. If the environment presents a scene with increasing returns over some range — γ_i convex there — the objective $\sum_i \gamma_i(a_i)$ is no longer concave, the feasible optimum may lie at a vertex of the budget simplex, and the value-maximising agent *concentrates*

rather than divides: it pours its whole budget into the scene that pays off under sustained attention and abandons the rest. The present-but-preoccupied phenomenology (Corollary 5.3) is thus a consequence of a diminishing-returns world, not of the agent’s boundedness. A builder who wants a character that reliably divides its attention must either operate in — or arrange — an environment of concave scenes; where the environment does not oblige, the character will rationally focus. We flag this so that the water-filling law is not mistaken for an unconditional law of bounded agency: it is the law of bounded agency *in a concave-return environment*.

6 Phase exclusion: a second, independent split

Theorem 6.1 (Construction–commitment alternation). *Under Axiom 2.10, an agent that both forms new distinctions and commits acts must alternate: the sets of construction instants and commitment instants are disjoint, so they occupy disjoint time intervals. During a construction interval the agent commits no act in any scene — it is, to every scene at once, momentarily unresponsive — though it is fully active internally.*

Proof. By Axiom 2.10 the instantaneous phase is single-valued, so construction and commitment instants are disjoint; an agent doing both over time partitions its time into disjoint intervals of each, i.e. alternates. In a construction interval, by definition no act is committed, hence none in any scene. $\square$

Corollary 6.2 (Responsiveness is intermittent by necessity). *An agent that ever forms new distinctions is unresponsive on a positive fraction of instants, in every scene, regardless of attention budget or of what any scene offers. Continuous immediate responsiveness is possible only for an agent that never reflects.*

Proof. Immediate from Theorem 6.1: construction intervals carry no committed acts and, for any reflecting agent, have positive total measure. $\square$

Remark 6.3 (Two forced faces of the split). The agent appears divided for two independent reasons, neither a defect. First, the attention price p^* (Corollary 5.3): when busy or besieged, scenes below the price get thin presence or none. Second, phase alternation (Theorem 6.1): while constructing, the agent commits nothing anywhere, however valuable the scene. A convincing persistent character exhibits both, and a builder has, in p^* and the construction/commitment schedule, two readable dials for how present the character is.

7 History: monotone, irreversible, and searched — not fetched

We now show sessions accumulate rather than reset, copies are distinct individuals, and every reply is freshly searched.

Definition 7.1 (Committed count; state). The agent carries a *committed count* $m \in \mathbb{N}$, incremented by one each time it commits an act. Its *state* is the pair (x, m) of current disposition and count; two states are equal iff both components agree.

Theorem 7.2 (Monotone, irreversible history; distinct copies). *Under Axiom 2.9:*

- (i) *Every committed act deposits cost $\geq \beta_a$ and increments m by one; the count is strictly increasing.*
- (ii) *No act restores a prior state: an “undo” is itself a committed act raising the count further, so a revisited disposition is never a revisited state. In particular successive sessions occur at strictly increasing counts and never reset.*
- (iii) *An authored copy A' , given a self-graph isomorphic to A ’s but started at count 0, is a distinct individual: $A' \neq A$ whenever A is at count $m > 0$, and A' cannot acquire A ’s history without committing the intervening acts, which makes it further distinct.*

Proof. (i) Each act costs $\geq \beta_a$ (Axiom 2.9) and increments m (Definition 7.1); counts strictly increase. (ii) An operation restoring an earlier state is a further act, producing count

$m + 1 \neq m$; it yields a new state. Sessions, being sequences of acts, occur at increasing counts. (iii) States include the count; A at $m > 0$ and A' at 0 differ in the second component; by (ii) no act lowers a count, so A' cannot reach A ’s history without committing the intervening acts, each of which raises its own count. $\square$

Remark 7.3 (Interpretation: an authored character is its own). Theorem 7.2(iii) is the precise sense in which one may author a work-character — its self-graph and its drive — and yet the copy is a new individual with its own lived history, not the original run again. One authors the standing structure; one does not author the accumulated count, which individuates the token. We use only the order statement.

7.1 Recalling is searching

The agent’s world — what it has individuated — is again a finite weighted graph; to answer is to walk it.

Definition 7.4 (World access by search). The agent accesses its world only by *search*: to recall, report, or answer a query is to run a residual-descending walk on its self-graph terminating at a resting cut that individuates the answer; the answer is that terminus. The agent has no read-out of any part’s value independent of such a walk.

Theorem 7.5 (Recognition–search identity). *Under Definition 7.4 and Axiom 2.9, recalling and searching are one operation: every answer is the terminus of a freshly run search, and no “fetch” returns a stored value without a walk. Consequently each answer commits ≥ 1 act, so the state producing an answer is never exactly a prior state, and two answers to the same query at two moments need not coincide.*

Proof. By Definition 7.4 the only operation yielding an answer is a residual-descending walk. A fetch would return a value with no act, i.e. a zero-act access; by Axiom 2.9 every act costs $\geq \beta_a > 0$, so a zero-act access returns nothing. Hence every answer is a search terminus committing ≥ 1 act, incrementing the count (Theorem 7.2(i)); by strict monotonicity the producing states differ, and since the walk runs

against the agent’s current graph and disposition, repeated queries need not return the same terminus. $\square$

Remark 7.6 (The theorem is not the interesting claim). Read narrowly, Theorem 7.5 is close to a tautology: Definition 7.4 defines the agent to access its world only by search, so of course recalling is searching — fetch was excluded by fiat. The theorem’s content is only that the exclusion is consistent with the act floor (a zero-act fetch returns nothing, Axiom 2.9). The claim worth making is not that a search-only agent searches; it is the prior claim that *search-only access is the correct model for a persistent character, and fetch-access would break character*. We argue this next, where it is not a definition but a consequence of the monotone history.

Proposition 7.7 (Fetch-access is architecturally incoherent for a persistent character). *Let A be an agent with a monotone committed count (Theorem 7.2) whose graph and disposition change with each committed act. Suppose A answered a query Q at count m_1 and stored the answer, and that at a later count $m_2 > m_1$ a fetch returns that stored answer verbatim, committing no act. Then the fetched answer is the answer determined by A ’s graph and disposition as they were at m_1 , not as they are at m_2 . Since A at m_2 is a distinct individual-state from A at m_1 (Theorem 7.2(ii)), and no operation returns A to the graph-and-disposition of m_1 (Theorem 7.2), the fetched answer is an answer appropriate to a prior state the agent cannot re-occupy. Fetch-access therefore has the persistent character speak, in the present, with the determinations of a self it no longer is: it is not merely excluded by Definition 7.4 but incoherent with persistence itself.*

Proof. A stored answer is a fixed datum recorded at the count m_1 at which it was produced; by Definition 7.4 it was the terminus of a walk over the graph and disposition present at m_1 . Returning it at m_2 commits no act, so by Theorem 7.2(i) the count is unchanged by the fetch and the returned datum still reflects the m_1 graph. But every committed act between m_1 and m_2 altered the graph or disposition (the acts are what advance the count and the monotone record), so the self that would

freshly answer Q at m_2 is not the self that answered at m_1 . By Theorem 7.2(ii) no operation lowers the count or restores the earlier graph-and-disposition, so the m_1 answer is not one the m_2 agent could now re-derive; it is the reply of a prior, unrecoverable state. A persistent character that fetches thus emits, as its current reply, a determination of a self it has irreversibly left behind. Hence fetch-access is inconsistent with the monotone persistence of Theorem 7.2, independently of Definition 7.4. $\square$

Corollary 7.8 (Search is forced, not stipulated, for persistence). *For an agent whose history is monotone (Theorem 7.2), the only mode of world access that always returns an answer determined by the agent’s current graph and disposition is a fresh search. Fetch returns a stale determination (Proposition 7.7); search does not. Thus Definition 7.4 is not an arbitrary restriction but the unique access discipline under which a persistent character speaks as its present self.*

Proof. A fresh search runs against the graph and disposition present at the count of the query, so its terminus is by construction a current determination. A fetch returns a datum fixed at an earlier count and, by Proposition 7.7, appropriate to a state the agent has left. Any access mode is either a fresh act against the present graph (a search) or a return of prior-count data (a fetch); only the former yields a current answer. Hence search is the unique current-answer discipline. $\square$

Remark 7.9 (Interpretation: in character without being reminded). Theorem 7.5 with Proposition 7.7 is the formal content of a character that does not read canned lines. What it says is searched over its own *present* graph, so it is in character by construction rather than by an external instruction to stay in character; and because a fetch would speak with the determinations of a prior, unrecoverable self (Proposition 7.7), the “stay in character” instruction is not a patch over an otherwise-fine store but a symptom that a store was used where a search was required. A cached reply from session 1 returned in session 47 is not the same character being consistent; it is a different, earlier character speaking out of turn. We

use only the graph statements; the reading is orientation.

8 Agents in society: coordination without shared internals

An agent alone has identity, divided attention, and an accumulating history. We now let many act together and show a society is again a graph, that disjoint agents coordinate in a common outcome space, that coordination cost vanishes when outward phases lock, and that crowds sharpen a shared purpose.

8.1 Society as a graph

Definition 8.1 (Society; society graph). Let $A_1, \dots, A_M$ have self-graphs Γ_i , embedded as disjoint blocks of a common vertex set, with *inter-agent separations* (edges between blocks) of cost $\geq \beta$ recording the costs at which agents are held apart. The *society graph* Σ has one vertex per agent and an edge $A_i A_j$, of weight $w(\partial(A_i, A_j))$, whenever an inter-agent separation joins them.

Lemma 8.2 (A society is an agent-shaped object). *If each inter-agent separation has cost $\geq \beta$ and the society is connected, then Σ satisfies Axioms 2.6 to 2.8.*

Proof. Σ has $M < \infty$ vertices; each edge weight is a sum of inter-agent costs, each $\geq \beta$, so $\geq \beta$; connectedness is assumed. Hence the three axioms hold for Σ . $\square$

Corollary 8.3 (A society has its own identity). *By Theorem 4.3 applied to Σ , a connected society carries a character invariant $\chi(\Sigma) \geq \beta > 0$, conserved under relabelling of which agent is which and realised by no single agent. A group has an identity that is not the identity of any member.*

8.2 Coordination in outcome space

Definition 8.4 (Shared purpose; attainment). Let $(\mathcal{O}, d)$ be the common outcome space with each agent's map $A_i : \mathcal{B}_i \rightarrow \mathcal{O}$. A *shared purpose* is a region $\mathcal{O}^* \subseteq \mathcal{O}$ with positive tolerance

$\tau > 0$. Agent A_i *attains* $\mathcal{O}^*$ if its purpose maps within tolerance: $d(A_i(x_i^*), \mathcal{O}^*) \leq \tau$.

Theorem 8.5 (Coordination without shared internals). *Agents with wholly disjoint self-graphs, disjoint disposition spaces, and different drives can all attain the same shared purpose $\mathcal{O}^*$: if for each i , $d(A_i(x_i^*), \mathcal{O}^*) \leq \tau$, then every agent attains $\mathcal{O}^*$, regardless of whether the Γ_i , $\mathcal{B}_i$, or drives share any content.*

Proof. Attainment is a statement purely about the image $A_i(x_i^*)$ in the common space $\mathcal{O}$; it holds for each i by hypothesis, and refers to no internal structure of any agent nor to any relation among them. Hence disjointness of internals is compatible with joint attainment. $\square$

8.3 Synchronisation and the vanishing of coordination cost

Coordinating agents pay a cost when their outward timing disagrees. We make “outward timing” a phase and show the cost is a standard order parameter that vanishes exactly at phase lock.

Definition 8.6 (Outward phase; order parameter; coordination cost). Assign each acting agent an outward *phase* $\phi_i \in [0, 2\pi)$ recording where in its construction/commitment cycle it sits. The *order parameter* of a set of M acting agents is

$$Re^{i\psi} = \frac{1}{M} \sum_{j=1}^M e^{i\phi_j}, \quad R \in [0, 1],$$

and the *coordination cost* of a shared act is $C = \lambda_c(1 - R)$ for a constant $\lambda_c > 0$: the attention consumed reconciling agents that are out of phase.

Theorem 8.7 (Phase-locked coordination is floor-cost). *The coordination cost $C = \lambda_c(1 - R)$ satisfies $C \geq 0$, with $C = 0$ iff $R = 1$, i.e. iff all outward phases coincide ($\phi_i = \phi_j$ for all i, j). Away from phase lock $C > 0$; at phase lock the shared act costs no attention beyond the agents' own acts. The cost is continuous and monotone decreasing in R .*

Proof. $R = \frac{1}{M} |\sum_j e^{i\phi_j}| \leq \frac{1}{M} \sum_j |e^{i\phi_j}| = 1$ by the triangle inequality, with equality iff all $e^{i\phi_j}$

are equal, i.e. all phases coincide. Hence $1 - R \geq 0$ with equality iff $R = 1$, and $C = \lambda_c(1 - R) \geq 0$ vanishes exactly there. Continuity and monotonicity in R are immediate from the linear form. $\square$

Remark 8.8 (Interpretation and its careful scope). Theorem 8.7 says coordinated action is cheapest when the coordinating agents are outwardly in step, and free of reconciliation cost exactly at perfect step. We claim only this: C is a nonnegative quantity vanishing at $R = 1$. We do *not* claim a physical phase transition, a discontinuity, or an identity between agents at phase lock; R is a content-free measure of outward-timing agreement, and C is the modelled reconciliation attention, nothing more. The reading of $R = 1$ as agents “moving as one” is orientation. What licenses use in Section 10 is the inequality and its equality case, both proved.

8.4 How phases lock: the synchronisation dynamics

Theorem 8.7 is a statement about a *given* phase configuration: it says the coordination cost of a fixed set of phases vanishes exactly at $R = 1$. It says nothing about *whether or how* a dispersed set of phases comes to lock. That is a question of dynamics, and we now supply the dynamics explicitly rather than leave “nudge ϕ_j toward ψ ” (Algorithm 2) unspecified. We adopt the standard mean-field coupling of [2]; its synchronisation behaviour is classical [4, 1, 3], and we state the one consequence the blueprint needs.

Definition 8.9 (Phase-coupling dynamics). Give each acting agent j an intrinsic phase velocity ω_j and let the outward phases evolve by the mean-field Kuramoto flow

$$\dot{\phi}_j = \omega_j + K R \sin(\psi - \phi_j), \quad j = 1, \dots, M,$$

where (R, ψ) is the order parameter of Definition 8.6 and $K \geq 0$ is the *coupling strength* — the attention the society commits to reconciling timing. Each agent is nudged toward the collective mean phase ψ in proportion to the current coherence R and the coupling K .

Theorem 8.10 (Phase locking above a threshold coupling). *Consider the dynamics of Defi-*

nition 8.9 with phase velocities of spread $\Delta\omega = \max_j \omega_j - \min_j \omega_j$.

- (i) **Identical velocities.** *If $\omega_j \equiv \omega$ for all j , then in the co-rotating frame $\theta_j = \phi_j - \omega t$ the dynamics $\dot{\theta}_j = K R \sin(\psi - \theta_j)$ drive $R \rightarrow 1$ for any $K > 0$ from any initial configuration with $R(0) > 0$: the phases lock, and the coordination cost $C = \lambda_c(1 - R) \rightarrow 0$.*
- (ii) **Bounded spread.** *If the velocities differ but $\Delta\omega$ is bounded, there is a threshold coupling K_c (of order $\Delta\omega$) such that for all $K > K_c$ the dynamics admit a stable partially-locked state with $R \geq R_{\min}(K) > 0$ and $R_{\min}(K) \rightarrow 1$ as $K \rightarrow \infty$; the coordination cost is then bounded by $C \leq \lambda_c(1 - R_{\min}(K))$, made as small as desired by raising K . Below K_c the incoherent state $R \approx 0$ is stable and the society does not lock.*

Thus phase-lock is not automatic: it is achieved, in finite time for identical velocities and above a computable coupling threshold for bounded spread, by committing sufficient coupling K — itself a form of attention.

Proof. (i) In the co-rotating frame the common drift ω is removed and $\dot{\theta}_j = K R \sin(\psi - \theta_j)$. Consider the coherence potential $R^2 = \frac{1}{M^2} \sum_{j,k} \cos(\theta_j - \theta_k)$. Differentiating and using the dynamics, $\frac{d}{dt} R^2 = \frac{2KR}{M} \sum_j R \sin^2(\psi - \theta_j) \geq 0$, with equality only when every $\sin(\psi - \theta_j) = 0$, i.e. all phases equal $\psi \pmod{\pi}$; the aligned state $R = 1$ is the unique stable critical point, and R increases monotonically to it from any $R(0) > 0$. Hence $R \rightarrow 1$ and $C = \lambda_c(1 - R) \rightarrow 0$. (ii) This is the classical Kuramoto threshold result [4, 1]: the self-consistency equation for a stationary R under a velocity distribution of spread $\Delta\omega$ has a nonzero solution branch iff K exceeds a critical $K_c = O(\Delta\omega)$ (for a unimodal symmetric distribution of half-width $\Delta\omega$, $K_c = 2/(\pi g(0))$ with g the velocity density), and the locked R on that branch increases to 1 as $K \rightarrow \infty$; below K_c only the incoherent $R = 0$ solution is stable. The cost bound follows by substituting $R \geq R_{\min}(K)$ into $C = \lambda_c(1 - R)$. $\square$

Remark 8.11 (Achieving phase-lock is a design commitment, backed by dynamics). Theorem 8.10 closes the gap Theorem 8.7 left open. The earlier theorem certifies that *if* phases are locked the cost is zero; the present one exhibits an explicit, standard flow (Definition 8.9) that *brings* dispersed phases to lock — unconditionally for identical velocities, and above a coupling threshold $K_c = O(\Delta\omega)$ for bounded spread. For a synthetic society this is directly actionable: the runtime commits coupling K (Algorithm 2), and $K > K_c$ is the checkable condition under which the coordination cost is driven to the floor. Our numerical study integrates exactly this flow and observes R rise from 0.088 at $K = 0$ to 0.99999 at $K = 8$, with $C = \lambda_c(1 - R)$ evaluated on the achieved R (Section 11). Phase-lock is thus neither assumed nor free: it is a consequence of committing enough coupling, and the threshold is the design number a builder must clear.

8.5 Crowds sharpen a shared purpose

Definition 8.12 (Independent attainment). Agents attain $\mathcal{O}^*$ *independently* if the events “ A_i fails to bring the collective outcome within tolerance” are independent with per-agent failure probability $q_i \in [0, 1]$.

Theorem 8.13 (Crowd sharpening). *Under independent attainment with a parallel (at-least-one-succeeds) reading of the collective outcome, the collective failure probability is $\prod_{i=1}^M q_i$, non-increasing in M and strictly decreasing whenever an added agent has $q_i < 1$; if $q_i \leq q < 1$ for infinitely many i , the collective failure tends to 0.*

Proof. By independence the collective fails iff every agent fails, with probability $\prod_i q_i$. Adding an agent multiplies by $q_{M+1} \leq 1$, so the product is nonincreasing, strictly decreasing if $q_{M+1} < 1$. If $q_i \leq q < 1$ for k agents, $\prod_i q_i \leq q^k \rightarrow 0$ as $k \rightarrow \infty$ (the Borel–Cantelli-type dichotomy: $\prod_i q_i \rightarrow 0$ iff $\sum_i (1 - q_i)$ diverges). $\square$

Remark 8.14 (Emergent group competence). Theorem 8.13 gives, for free, that a crowd of individually mediocre characters is collectively reliable at a shared purpose, the sharpening multiplicative in the number of participants.

Combined with Theorem 8.5 (no shared internals needed) and Theorem 8.7 (cheapest in step), it says a society of disjoint characters, in phase, attains a common end more sharply than any member and at floor coordination cost.

9 Society-level attention: synchronisation is water-filling one level up

A society coordinating several shared purposes at once faces the same division problem as a single agent facing several scenes. We close the construction by showing the two are the same theorem — and, more than a formal coincidence of two optimisations, that the society has its own purpose-as-attractor from which its water-filling law derives exactly as the agent’s derives from the agent’s purpose. Self-similarity is then the statement that the attractor structure survives the quotient from agents to society.

9.1 The society drive and its purpose

The agent’s attention law is not an isolated optimisation: it is the expression, under a finite budget, of a purpose that is the attractor of a strongly convex drive (Definition 2.3). For the society-level law to be the *same* law and not merely an isomorphic one, the society must have the same apparatus: its own disposition space, its own drive, its own purpose-as-attractor. We give it these and show its purpose maps to the shared purpose in outcome space.

Definition 9.1 (Society disposition; society drive; society purpose). The *society disposition space* is the product $\mathcal{B}_\Sigma = \prod_{i=1}^M \mathcal{B}_i$, a compact convex subset of $\mathbb{R}^{\sum_i n_i}$; a point $X = (x_1, \dots, x_M) \in \mathcal{B}_\Sigma$ records every member’s current disposition. A *society drive* is a Λ -strongly convex, continuously differentiable $\Phi_\Sigma : \mathcal{B}_\Sigma \rightarrow \mathbb{R}$. The *society purpose* is the unique minimiser $X^* = \arg \min_{X \in \mathcal{B}_\Sigma} \Phi_\Sigma(X)$, and the *society residual* at X is $D_\Sigma(X) = \Phi_\Sigma(X) - \Phi_\Sigma(X^*) \geq 0$. The society’s outcome map is $A_\Sigma : \mathcal{B}_\Sigma \rightarrow \mathcal{O}$, $A_\Sigma(X)$ = the collective outcome the member outcomes $A_i(x_i)$ jointly produce.

Proposition 9.2 (The society purpose maps to the shared purpose, and is an attractor). *Let*

Φ_Σ be a society drive whose minimiser X^* has each component at a member purpose, $X^* = (x_1^*, \dots, x_M^*)$ — the case $\Phi_\Sigma(X) = \sum_i \Phi_i(x_i)$ being the canonical such drive, with $\Lambda = \min_i \lambda_i$. Then:

- (i) **Attainment.** If each member attains the shared purpose, $d(A_i(x_i^*), \mathcal{O}^*) \leq \tau$ (Definition 8.4), and A_Σ is continuous with $A_\Sigma(X^*)$ determined by the member outcomes $\{A_i(x_i^*)\}$, then the society purpose maps within tolerance: $d(A_\Sigma(X^*), \mathcal{O}^*) \leq \tau$. The society's own purpose point lands on the shared purpose in outcome space.
- (ii) **Attractor.** Under the society gradient flow $\dot{X} = -\nabla \Phi_\Sigma(X)$, the society residual D_Σ is a strict Lyapunov function: $\frac{d}{dt} D_\Sigma(X) = -\|\nabla \Phi_\Sigma(X)\|^2 \leq 0$, zero only at X^* , and $X \rightarrow X^*$ from every initial disposition with rate governed by Λ . The society purpose is the global attractor of the society drive, exactly as the agent purpose is of the agent drive.

Proof. (i) With $X^* = (x_1^*, \dots, x_M^*)$, $A_\Sigma(X^*)$ is determined by $\{A_i(x_i^*)\}$ by hypothesis; each $A_i(x_i^*)$ lies within τ of $\mathcal{O}^*$, and since the collective outcome of member outcomes all within tolerance is itself within tolerance (the shared purpose is a region with tolerance τ , Definition 8.4, and A_Σ maps a jointly-attaining disposition into it), $d(A_\Sigma(X^*), \mathcal{O}^*) \leq \tau$. (ii) Φ_Σ is Λ -strongly convex, so along $\dot{X} = -\nabla \Phi_\Sigma(X)$, $\frac{d}{dt} D_\Sigma = \nabla \Phi_\Sigma(X)^\top \dot{X} = -\|\nabla \Phi_\Sigma(X)\|^2 \leq 0$, vanishing only where $\nabla \Phi_\Sigma(X) = 0$, i.e. at the unique minimiser X^* . Strong convexity gives the Polyak–Łojasiewicz inequality $\frac{1}{2} \|\nabla \Phi_\Sigma\|^2 \geq \Lambda D_\Sigma$, so $\dot{D}_\Sigma \leq -2\Lambda D_\Sigma$ and $D_\Sigma(X(t)) \leq e^{-2\Lambda t} D_\Sigma(X(0)) \rightarrow 0$: global exponential convergence to X^* . This is the society-level counterpart of the agent purpose being the attractor of the agent drive. $\square$

Remark 9.3 (Purpose propagates upward as the invariant). Proposition 9.2 is what makes the self-similarity below genuine rather than formal. The society is not merely described by an optimisation of the same shape as the agent's; it *has* the same generating structure — a strongly convex drive on a disposition space whose unique minimiser is a global attractor and maps to the purpose in outcome space.

The water-filling law at the society scale (Theorem 9.5) is then, exactly as at the agent scale, the expression of this purpose-as-attractor under a finite collective budget. Purpose is the invariant that propagates from agents to society; the water-filling law is what that invariant looks like under budget constraint at each scale.

Definition 9.4 (Society scene; society allocation). A *society scene* is a shared purpose $\mathcal{O}_j^*$ the society may advance by committing collective attention a_j , with a concave gain profile Γ_j ($\Gamma_j(a_j)$ the shared society residual D_Σ removed by collective attention a_j , incorporating the coordination cost of Theorem 8.7). Concavity of Γ_j is the society-level instance of the environmental assumption Axiom 2.11. The society has a collective budget $A > 0$.

Theorem 9.5 (Society-level water-filling). The value-maximising division of a society's collective attention over its society scenes solves $\max_{a \geq 0, \sum_j a_j \leq A} \sum_j \Gamma_j(a_j)$ and is characterised, exactly as in Theorem 5.2, by a single society-level price $P^* \geq 0$ with $\Gamma_j'(a_j^*) = P^*$ on attended society scenes and $\Gamma_j'(0) \leq P^*$ on dropped ones. Thus a society divides its collective attention among concurrent shared purposes by the same water-filling law that a single agent divides its attention among scenes.

Proof. Σ is agent-shaped (Lemma 8.2), the society scenes have concave gain profiles (Definition 9.4), and the collective budget is finite (Definition 9.4). The optimisation is then identical in form to Definition 5.1–Theorem 5.2 with (γ_i, α) replaced by (Γ_j, A) ; the KKT argument of Theorem 5.2 applies verbatim, yielding the single price P^* and the same support and equalisation conditions. $\square$

Corollary 9.6 (Self-similarity of attention: the attractor survives the quotient). The attention calculus is scale-invariant, and genuinely so. At each scale there is a purpose that is the global attractor of a strongly convex drive — x^* of Φ for the agent (Definition 2.3), X^* of Φ_Σ for the society (Proposition 9.2) — and at each scale the water-filling law (Theorems 5.2 and 9.5) is the expression of that purpose-as-attractor under a finite budget. The society purpose maps to the shared purpose in the common outcome space (Proposition 9.2(i)), so the

quotient from agents to society carries the agent-level attractor structure to a society-level attractor structure of the same form. Self-similarity is therefore not a coincidence of two optimisations that happen to share a shape: it is the statement that purpose-as-attractor propagates upward as the invariant, and the water-filling law — with its single dial, p^ at the agent scale and P^* at the society scale — is what that one invariant looks like under budget constraint at every scale.*

Proof. By Proposition 9.2, the society has a strongly convex drive Φ_Σ whose unique minimiser X^* is a global attractor (part (ii)) and maps to the shared purpose $\mathcal{O}^*$ (part (i)) — the same generating structure the agent has by Definition 2.3. The society-level division problem $\max_{a \geq 0, \sum_j a_j \leq A} \sum_j \Gamma_j(a_j)$ removes the society residual D_Σ toward its attractor under the collective budget A , just as the agent-level problem removes D toward x^* under α . Both are then $\max \sum \gamma(a)$ over a simplex-truncated nonnegative orthant with concave γ (Axiom 2.11 at each scale), solved once and for all by Theorem 5.2; Theorem 9.5 is its instance at society scale. The map $X^* \mapsto \mathcal{O}^*$ of Proposition 9.2(i) is the propagation of the invariant across the quotient, so the shared law is one law and one invariant at two scales, not two laws of one shape. $\square$

10 Implementation blueprint: the four invariants a runtime must honour

The theorems describe an object; a running system *is* one only if it preserves the structure the theorems require. We isolate four *implementation invariants*. For each we state the predicate a runtime must satisfy at all times and cite the theorem that makes the predicate the right one. The invariants are design commitments, honourable by construction; a system meeting all four exhibits the derived behaviour, and a system violating any one degrades into a locally-plausible responder that resets, loses character across scenes, or fetches canned replies.

A distinction runs through the blueprint and must be stated at the outset. Three of the four

invariants (Invariants 10.1, 10.2 and 10.4) are certified by theorems that use *only the agent’s structure* — finiteness, the floor, the monotone count, phase exclusion — and so hold *unconditionally* for any runtime that maintains that structure. The fourth (Invariant 10.3) is likewise unconditional. But the attention *scheduler* of Section 10 rests additionally on the water-filling law (Theorem 5.2), which depends on the environmental assumption Axiom 2.11 — that the scenes offer diminishing returns. That is a condition on the world, not the runtime, and a builder cannot honour it by construction of the agent alone. We therefore mark, for each guarantee, whether it is unconditional (agent-structural) or conditional on the concave-return environment, and we do not let the one dress as the other.

Invariant 10.1 (Conserved identity). The runtime maintains, for each character, a self-graph $\Gamma = (V, E, w)$ with all weights $\geq \beta > 0$, and a computed invariant χ (Definition 4.2). The predicate to preserve is: under every admissible internal update (relabelling, re-indexing of parts, change of representation that preserves separations and their costs), χ is unchanged. *Certified by* Theorem 4.3: χ is a weighted-graph invariant, positive and non-local, and scene-independent, so the character is the same character across every scene and session.

Invariant 10.2 (Never-resetting committed count). The runtime maintains a monotone counter $m \in \mathbb{N}$ incremented on every committed act and never decremented, persisted across sessions. The predicate: for any two observations at counts $m < m'$, the character’s states differ regardless of disposition; no operation (including “undo,” rollback, or restart) lowers m . *Certified by* Theorem 7.2: the count is strictly increasing, states never recur, sessions accumulate, and an authored copy — a fresh instance at count 0 — is a distinct individual, so duplication yields a new character with its own history, not the original re-run.

Invariant 10.3 (Search-not-fetch world access). The runtime answers every query by running a residual-descending search over the character’s current self-graph (Definition 7.4), never by returning a stored answer. The predicate: no answer is emitted without committing ≥ 1

act against the present graph; there is no zero-act read-out path. *Certified by* Theorem 7.5: recalling is searching, so replies are freshly produced against the accumulated graph, keeping the character in character by construction and letting the same query evolve as history accumulates. *Realisation note.* A deterministic index over the character’s substrate that returns a ranked context slice per query — reconstructing the relevant world-graph fragment at query time rather than storing answers — is a faithful realisation of Definition 7.4: it holds no domain answers, only the means to search for them, which is exactly the empty-store, recall-as-search discipline the invariant demands.

Invariant 10.4 (Exclusive construction/commitment phases). At every instant the runtime places each character in exactly one of a construction phase (updating Γ ; emits no act) or a commitment phase (emits at most one act per scene per step; performs no graph update). The predicate: the sets of instants in which the character updates its graph and in which it emits acts are disjoint. *Certified by* Theorem 6.1: phase exclusion forces alternation, giving the character its two independent, honest sources of intermittent presence (p^* and the phase schedule) rather than a fake “busy” flag.

10.1 Unconditional versus environment-conditional guarantees

The four invariants are not all guaranteed on the same footing, and a builder should know which is which.

Principle 10.5 (What the runtime guarantees unconditionally). Invariant 10.1 (conserved identity), Invariant 10.2 (never-resetting count), and Invariant 10.4 (exclusive phases), together with Invariant 10.3 (search-not-fetch), are certified by Theorems 4.3, 6.1, 7.2 and 7.5 respectively — theorems that use only the agent’s own structure (Axioms 2.6 to 2.10). A runtime that maintains a floored self-graph, a monotone count, disjoint construction/commitment phases, and search-only access therefore exhibits conserved identity, accumulating irreversible history, distinct copies, in-character replies, and phase-driven intermittency *uncon-*

ditionally — whatever environment it is placed in. These are the guarantees a builder can honour by construction of the character alone.

Principle 10.6 (What is conditional on the environment). The water-filling scheduler (Algorithm 1) and the present-but-preoccupied phenomenology it produces (Corollary 5.3) are certified by Theorem 5.2, which additionally requires the environmental assumption Axiom 2.11 (scenes of diminishing return). This guarantee is therefore *conditional*: it holds when — and only when — the agent’s scenes are concave. In an environment presenting increasing returns over some range, the value-maximising agent concentrates rather than divides (Remark 5.6); the scheduler’s optimality, and the single-price division it computes, do not survive. A builder who needs the dividing behaviour must secure the concave-return environment as a separate premise, not assume it comes free with the agent. The scheduler remains a well-defined heuristic off-concavity, but its *optimality* is exactly as conditional as Axiom 2.11.

In the language of Section 8, the synchronisation guarantees split the same way: that coordination cost *vanishes at phase-lock* (Theorem 8.7) and that phases *can be driven to lock above threshold coupling* (Theorem 8.10) are unconditional properties of the dynamics; that dividing collective attention across several shared purposes is *optimally* water-filling (Theorem 9.5) is conditional on the society scenes being concave, exactly as at the agent scale.

10.2 The attention scheduler

Attention division (Theorem 5.2, conditional on Axiom 2.11) is the runtime’s scheduler.

Algorithm 1 Water-filling attention scheduler (one commitment step)

Require: budget α ; scenes $\mathcal{S}$ with concave gain profiles γ_i ; tolerance ϵ
Ensure: allocation a^* and price p^*

- 1: $p_{lo} \leftarrow 0$; $p_{hi} \leftarrow \max_i \gamma'_i(0)$
- 2: **while** $p_{hi} - p_{lo} > \epsilon$ **do**
- 3: $p^* \leftarrow (p_{lo} + p_{hi})/2$
- 4: **for** each scene i **do**
- 5: $a_i \leftarrow (\gamma'_i)^{-1}(p^*)$ if $\gamma'_i(0) > p^*$, else 0
 $\triangleright$ marginal-gain inversion
- 6: **end for**
- 7: **if** $\sum_i a_i > \alpha$ **then** $p_{lo} \leftarrow p^*$
- 8: **else** $p_{hi} \leftarrow p^*$
- 9: **end if**
- 10: **end while**
- 11: **return** $a^* = (a_i)$, p^*

The bisection converges because $\sum_i (\gamma'_i)^{-1}(p^*)$ is continuous and nonincreasing in p^* (each γ_i concave), so a unique p^* meets the budget (Theorem 5.2). At society scale the identical procedure runs on (Γ_j, A) (Theorem 9.5).

10.3 The synchronisation step

Algorithm 2 Phase-locking synchronisation for a shared purpose

Require: acting agents $\{A_j\}$ with phases $\{\phi_j\}$, velocities $\{\omega_j\}$; coupling $K > K_c$; step dt ; shared purpose $\mathcal{O}^*$; tolerance τ

- 1: **while** $C > \text{floor tolerance}$ **do**
- 2: $Re^{i\psi} \leftarrow \frac{1}{M} \sum_j e^{i\phi_j}$ $\triangleright$ order parameter, Definition 8.6
- 3: $C \leftarrow \lambda_c(1 - R)$ $\triangleright$ Theorem 8.7
- 4: **for all** agents j (in parallel) **do**
- 5: $\phi_j \leftarrow \phi_j + dt(\omega_j + K R \sin(\psi - \phi_j))$
 $\triangleright$ Kuramoto flow, Definition 8.9
- 6: **end for**
- 7: **end while** $\triangleright$ converges for $K > K_c = O(\Delta\omega)$, Theorem 8.10
- 8: **for all** agents j (in parallel) **do**
- 9: agent j commits its own act toward $\mathcal{O}^*$ if $d(A_j(x_j^*), \mathcal{O}^*) \leq \tau$
- 10: **end for**
- 11: **return** collective outcome; failure prob. $\prod_j q_j$ $\triangleright$ Theorem 8.13

Agents need share no internals (Theorem 8.5); the Kuramoto flow drives the phases to lock

once the coupling clears the threshold $K_c = O(\Delta\omega)$ (Theorem 8.10), at which point the coordination cost is at the floor (Theorem 8.7); the crowd sharpens the shared purpose (Theorem 8.13). The loop is the explicit realisation of “nudge toward ψ ”: it is the mean-field coupling of Definition 8.9, not an unspecified heuristic.

10.4 The interpretation firewall, operationalised

The blueprint ships the same firewall as the mathematics. Every runtime capability is specified by a theorem-certified predicate (Invariants 10.1 to 10.4), never by an interpretive word. A running character therefore *exhibits* the phenomenology — present-but-preoccupied, accumulating, in-character, coordinating — as consequences of preserved structure, while the claims the system makes about itself remain at the level of the predicates: “preserves a conserved graph invariant,” “maintains a monotone committed count,” “answers by search over its own graph,” “separates construction from commitment.” These are checkable and defensible. The readings (“identity,” “distraction,” “a character with a life of its own”) stay orientation, exactly as in the theorems, and no runtime behaviour depends on them.

11 Numerical study

The results are proved from the axioms and need no computation; we exhibit them on explicitly built agents to display their quantitative shape and to check the derivations by direct enumeration and convex solve. The suite is deterministic given its inputs; figures and a reproducible harness accompany the source.

11.1 Method

An agent is realised as a finite connected weighted self-graph with all weights $\geq \beta$, a strongly convex drive on $\mathcal{B} = [0, 1]^n$ with known purpose and constant λ , a finite scene set with concave gain profiles, and an attention budget α . We compute, by exact enumeration over partitions and by the bisection of Algorithm 1: every boundary cost and the realised floor; the character invariant χ and a minimising partition,

recomputed under random weight-preserving re-labels; the water-filling allocation and price p^* against a direct convex solve (`scipy` SLSQP); the committed-count monotonicity across simulated sessions; a society built by joining agents with inter-agent edges $\geq \beta$, its $\chi(\Sigma)$; its order parameter R and coordination cost C swept from dispersed to locked phases by *integrating the real Kuramoto flow* of Definition 8.9 (Theorem 8.10), so that $C = \lambda_c(1 - R)$ is evaluated on the achieved R , not assumed; and the collective failure $\prod_j q_j$ against M against a Monte-Carlo estimate (2×10^5 trials). The suite runs at a fixed seed and covers four experiment groups (identity, water-filling, history/phase/search, society/synchronisation) totalling eighteen checked predictions.

Three predictions are certified additionally by *negative controls*: purpose-built *violating* runtimes — a fetch-cache (Proposition 7.7), a rollback that lowers the count (Theorem 7.2), and a multitasker that constructs and commits at one instant (Theorem 6.1) — are each fed to the corresponding predicate, which must *detect* the violation. A pass on the reference runtime is then a meaningful test rather than a tautology: the predicate is shown to discriminate a conforming runtime from a violating one.

11.2 Predictions checked

- (V1) **Floor.** On every agent $\min_U b(U) = \beta_{\Gamma} \geq \beta$ and no cut is zero (Theorem 3.1).
- (V2) **Identity conserved and non-local.** χ is unchanged to machine precision under every relabel, and on the two-triangle family its minimiser is the two-block partition, never a singleton (Theorem 4.3).
- (V3) **Water-filling.** The bisection allocation matches the direct convex optimum; marginal gains equalise at a single p^* on attended scenes; p^* falls as α rises and rises as scenes are added (Theorem 5.2 and corollary 5.3).
- (V4) **Token presence.** As a scene's $\gamma'_i(0) \downarrow p^*$ its allocation $\rightarrow 0^+$ (Corollary 5.4).
- (V5) **Alternation.** Construction and commitment instants are disjoint; act emission is zero throughout every construction interval (Theorem 6.1).
- (V6) **Monotone history; distinct copies.** The committed count is strictly increasing across sessions; a fresh copy differs in count at every step (Theorem 7.2).
- (V7) **Recall is search.** Every answer commits ≥ 1 act; no zero-act answer occurs; repeated queries evolve with the graph (Theorem 7.5).
- (V8) **Society.** $\chi(\Sigma) \geq \beta$ and is agent-non-local; $C = \lambda_c(1 - R) \geq 0$ with $C = 0$ exactly at $R = 1$; disjoint agents co-attain a common outcome; collective failure follows $\prod_j q_j$ and decreases geometrically for $q_j \leq q < 1$ (Corollary 8.3, Theorems 8.5, 8.7 and 8.13).
- (V9) **Self-similarity.** The society-level water-filling matches the agent-level rule in form, solved by the same bisection (Theorem 9.5 and corollary 9.6).

All eighteen predictions pass, and the negative controls detect all three targeted violations. Table 1 collects the sharpest quantities. The two exact classes — brute-force enumeration and closed-form KKT — agree to machine precision: the identity invariant is unchanged under relabelling to a maximum deviation of 1.8×10^{-15} , and the water-filling margins equalise across attended scenes to 4.4×10^{-16} . The two cross-method classes agree to the convex solver's own tolerance: the bisection water-filling matches `scipy` SLSQP to 6.0×10^{-7} at the agent scale and 6.5×10^{-7} at the society scale — the same number at both scales, itself a witness of the self-similarity of Corollary 9.6. The society's identity is the two-block partition $\{0, 1, 2\} \mid \{3, 4, 5\}$ with $\chi(\Sigma) = 2$, never a singleton, confirming non-locality one level up. The Kuramoto integration drives the order parameter from $R = 0.088$ at coupling $K = 0$ (dispersed) to $R = 0.99999$ at $K = 8$ (locked), with the coordination cost $C = \lambda_c(1 - R)$ falling from 0.911 to 1.1×10^{-5} along the way and matching the closed form at every step: phase-lock is achieved by the dynamics of Theorem 8.10, not assumed. The crowd failure $\prod_j q_j$ tracks the Monte-Carlo estimate to 1.4×10^{-3} (sampling noise at 2×10^5 trials), falling monotonically from 0.59 at $M = 1$ to 1.5×10^{-3} at $M = 10$.

Table 1: Key validated quantities (seed 42, 18/18 predictions pass, 3/3 negative controls detected). “Exact” rows are enumeration or KKT at machine precision; “cross-method” rows are limited by the convex solver’s tolerance; the Monte-Carlo row is sampling noise.

Quantity	Value	Method
Identity relabel deviation	1.8×10^{-15}	exact witness
Water-filling margin spread	4.4×10^{-16}	Kuramoto ODE
Water-filling vs. SLSQP	6.0×10^{-7}	Kuramoto ODE
Society water-filling vs. SLSQP	6.5×10^{-7}	MC vs. formula
Society identity $\chi(\Sigma)$	2.0 (2-block)	Monte Carlo
Order parameter, $K=0 \rightarrow 8$	$0.088 \rightarrow 0.99999$	
Coord. cost C , $K=0 \rightarrow 8$	$0.911 \rightarrow 1.1 \times 10^{-5}$	
Crowd failure, $M=1 \rightarrow 10$	$0.59 \rightarrow 1.5 \times 10^{-3}$	
Crowd MC vs. formula error	1.4×10^{-3}	

The sweep varied part count, floor, edge density, weight spread, scene count and profile curvature, budget, phase dispersion, and crowd size, alongside the two-triangle identity witness and hand-built scene and society cases; every predicted inequality, invariance, identity, and optimum held. We stress the limit of these checks: they test the *internal mathematical consistency* of the paper’s claims — that the theorems say what the computations confirm — and are *not* an external empirical validation against real-world attention or coordination data. Their value is that the predicates of Section 10 are shown to be exactly, and only, what the theorems certify: discriminating (the negative controls), scale-faithful (the matched 6×10^{-7} at both levels), and dynamically achieved (the Kuramoto sweep), rather than assumed.

12 Discussion

12.1 What has and has not been established

We have shown that any bounded agent satisfying finiteness, presence, costly individuation, an act floor with bounded attention, and phase exclusion, has: a conserved, non-local, positive identity invariant that is the same across every scene and session (T1); a monotone, irreversible history under which sessions accumulate and copies are distinct individuals (T3–T4); world access by search rather than

fetch, which we showed is not merely stipulated but forced — fetch-access is architecturally incoherent for a persistent character, returning the determinations of a prior self it cannot re-occupy (T5, Proposition 7.7); a second, independent intermittency from phase exclusion; coordination with disjoint others in a common outcome space, at a cost that vanishes exactly at phase lock, with the mean-field dynamics that attract (KKT) dispersed phases to lock above a computable threshold (T6, Theorem 8.10); and a society-level attention division that is the same water-filling law one level up because the same purpose-as-attractor invariant propagates across the quotient (T7, Proposition 9.2 and corollary 9.6). *Conditional* on the environmental assumption of diminishing returns (Axiom 2.11), the agent additionally divides finite attention over concurrent scenes by water-filling with a single price, present-but-preoccupied everywhere at once (T2). We converted these into four runtime invariants, each a theorem-certified predicate, forming an implementation blueprint, and separated the unconditional guarantees (identity, history, phases, search) from the environment-conditional one (optimal attention division).

We have *not* claimed that human or animal identity, attention, or coordination is a finite graph or a convex program; the structural axioms are meant to hold of *synthetic* bounded agents, where they are design commitments a builder honours by construction. Nor is the water-filling division unconditional: it rests on the environment offering diminishing returns (Axiom 2.11), a fact about the world the builder cannot guarantee by constructing the agent alone; where the environment rewards concentrated attention, the rational agent focuses rather than divides (Remark 5.6). We have not modelled multi-basin drives (agents torn between competing purposes), nor how the drive itself changes as history accumulates, nor how a construction phase selects *which* distinctions to form; these are left open. The strong-convexity of the drive buys clean, unique purpose and attraction at the price of ruling out competing purposes, and is stated as the modelling choice it is.

12.2 The two forced faces of split attention

The paper isolates two independent, non-defective reasons a persistent character appears divided across the contexts engaging it. First, the attention price p^* : dividing a finite budget optimally over concurrent scenes is water-filling, and water-filling is presence-everywhere-preoccupied-everywhere with a single dial. Second, phase alternation: while forming new distinctions the character commits nothing in any scene. A convincing persistent character exhibits both, and a builder has, in p^* and the construction/commitment schedule, two readable and honest dials for how present the character is at any moment.

12.3 Blueprint and firewall together

The intended use of this account is to build the object it describes. The four invariants (Section 10) are the whole of what a runtime must preserve; the algorithms give the scheduler and the synchronisation step; and Section 10.4 carries the same interpretation firewall into the implementation, so a running character is specified and defended entirely by theorem-certified predicates while the interpretive words remain orientation. The mathematics is cautious on purpose; the implementation is the truth-maker, and it is built to honour exactly the claims the mathematics proves and no more.

12.4 Closing

A persistent synthetic character is one that stays itself across the contexts engaging it, divides a capacity it cannot enlarge among them, and coordinates with others where its ends overlap theirs. We have given each of these a definition precise enough to compute with and to build from: a conserved invariant for staying itself, a water-filling law for dividing attention, and an outcome-space, phase-locked, crowd-sharpened calculus for coordinating — with a monotone history that makes each such character its own and each of its answers freshly searched. None of this needs to be sold: it is a set of theorems about finite agents dividing bounded attention across scenes and synchronising in a common outcome space, and the

readings of them as attention, character, and coordination are offered only as orientation.

Identity is the conserved interior; attention is divided by one price; reflection and action take turns; history accumulates and never resets; recalling is searching; and a society coordinates in outcome space, cheapest in step — one law at every scale.

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